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Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n .

Example 2. $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
 $p(5) = 7$

Lemma 3. The generating function for $p(n)$ is given by

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \dots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \dots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n : j_k is the multiplicity of the part k . Hence the coefficient of x^n equals $p(n)$, and the first equality follows. $\square$

Definition 4 (Partitions of n into distinct parts). $\text{all_distinct_partitions}, \text{even_distinct_partitions}, \text{odd_distinct_partitions}$

Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned}\mathcal{P}(n) &= \left\{ S \subseteq \{1, \dots, n\} \mid \sum_{s \in S} s = n \right\}, \\ \mathcal{P}_{\text{even}}(n) &= \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod{2}\}, \\ \mathcal{P}_{\text{odd}}(n) &= \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod{2}\}.\end{aligned}$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty} (1 - x^k) = \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \dots < k_s} (-1)^s x^{k_1 + k_2 + \dots + k_s} = \sum_{n=0}^{\infty} c_n x^n$$

where $c_0 = 1$ and for $n \geq 1$ and $c_n = p_e(n) - p_o(n)$.

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1 - x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \dots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k \in K} (-x^k) = (-1)^{|K|} x^{\sum_{k \in K} k} = (-1)^s x^{k_1 + \dots + k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S \in \mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

$$c_n = \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ even}}} 1 - \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

□

Example 6. We compute $\prod_{i=1}^{20}(1-x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O(x^{20})$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler). *euler_pentagonal_number_theorem* $\prod_{i=1}^{\infty}(1-x^i) = 1 + \sum_{k=1}^{\infty}(-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}) = \sum_{k \in \mathbb{Z}}(-1)^k x^{(3k^2-k)/2}$.

Proof. The proof combines the previous expansion lemma with the closed-form computation of $p_e(n) - p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By the previous lemma,

$$\prod_{i=1}^{\infty}(1-x^i) = \sum_{n=0}^{\infty} c_n x^n,$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Step 2: Closed form for c_n . By Lemma ?? below,

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k & \text{if } n = (3k^2 - k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ (-1)^k & \text{if } n = (3k^2 + k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 0 & \text{otherwise.} \end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n \neq 0$ are exactly the (generalized) pentagonal numbers $n = (3k^2 \pm k)/2$ for $k \geq 1$, plus $n = 0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty}(1-x^i) = 1 + \sum_{k=1}^{\infty}(-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k \mapsto -k$. For $k \in \mathbb{Z}_{\geq 1}$, set $k' = -k$, so $k' \in \mathbb{Z}_{\leq -1}$ and

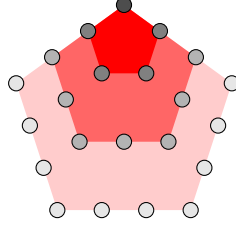
$$\frac{3(k')^2 - k'}{2} = \frac{3k^2 + k}{2}, \quad (-1)^k = (-1)^{|k'|} = (-1)^{k'}.$$

The $k = 0$ term gives $(-1)^0 x^0 = 1$, matching the leading 1. Combining the three contributions:

$$\sum_{k \in \mathbb{Z}_{\geq 1}} (-1)^k x^{(3k^2-k)/2} + (-1)^0 x^0 + \sum_{k' \in \mathbb{Z}_{\leq -1}} (-1)^{k'} x^{(3(k')^2-k')/2} = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

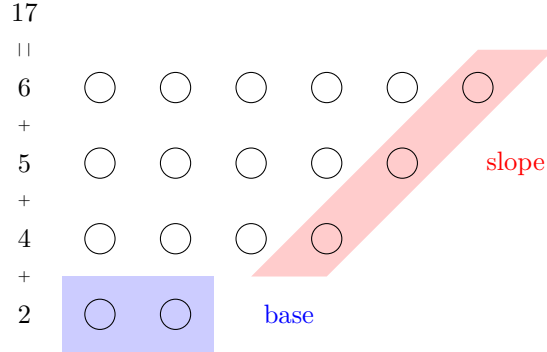
□

Remark 8 (Pentagonal numbers). $\frac{k}{\frac{3k^2-k}{2}} \left| \begin{array}{c} -3 \\ 15 \end{array} \right| \frac{-2}{7} \left| \begin{array}{c} -1 \\ 2 \end{array} \right| \frac{0}{0} \left| \begin{array}{c} 1 \\ 1 \end{array} \right| \frac{2}{5} \left| \begin{array}{c} 3 \\ 12 \end{array} \right|$



2.1 Proof

Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope). Consider a partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition. Let $\max(S)$ denote the largest element of S . We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 10. Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned} \mathcal{P}_\alpha(n) &= \{S \in \mathcal{P}(n) \mid b \leq s \text{ and } b \notin D, \text{ or if } b \leq s - 1\}, \\ \mathcal{P}_\beta(n) &= \{S \in \mathcal{P}(n) \mid b > s \text{ and } b \notin D, \text{ or if } b \geq s + 2\}, \\ \mathcal{P}_{\text{special}}(n) &= \{S \in \mathcal{P}(n) \mid b \in D \text{ and } |b| = |s| \text{ or } |b| = |s| + 1\}. \end{aligned}$$

Lemma 11. *For all natural numbers n the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.*

Proof. Throughout, let $S \in \mathcal{P}(n)$ be a partition with base b , slope s , slope set D . Recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff $(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff $(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and $(b = s \text{ or } b = s + 1)$.

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s + 1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ ($b \leq s - 1$) fails.
- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s + 1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s + 2$) fails since $b \leq s + 1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. □

Lemma 12. *For all natural numbers n the following holds*

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Proof. By the previous lemma the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case. If $S = \emptyset$ (which can occur only when $n = 0$), the base, slope, and slope set are not defined. We treat this case separately: there are no special, α , or β partitions of 0 in the sense above, so the statement reduces to $\mathcal{P}(0) = \emptyset \sqcup \emptyset \sqcup \emptyset$. TODO: confirm/define the convention for $n = 0$ — the natural reading is $\mathcal{P}(0) = \{\emptyset\}$ but no operation applies. The statement either needs $n \geq 1$ or a convention placing $\emptyset$ in one of the three sets.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We perform a trichotomy on b vs s :

Case 1: $b \leq s - 1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s + 1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s+1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s+1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s+2$. The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities (either $b \leq s+1$ or $b \geq s+2$; within $b \leq s+1$ we split into $b \leq s-1$, $b = s$, $b = s+1$). $\square$

Definition 13. For $k \in \mathbb{Z}_{\geq 1}$ define $S_{-k} = \{2k-1, 2k-2, \dots, k\}$.
For $k \in \mathbb{Z}_{\geq 0}$ define $S_k = \{2k, 2k-1, \dots, k\}$.

Lemma 14. If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.

If $n = \frac{3k^2-k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.

If $n = \frac{3k^2+k}{2}$ for some $k \in \mathbb{Z}_{\geq 0}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.

Proof. **TODO:** The statement and the definition of S_k for the upper-staircase family require reconciliation. As written, $S_k = \{2k, 2k-1, \dots, k\}$ has $k+1$ elements summing to $\sum_{i=k}^{2k} i = (3k^2+3k)/2 \neq (3k^2+k)/2$. The intended set is $\{k+1, k+2, \dots, 2k\}$ with k elements (and base $k+1$, slope k). Also the case $k=0$ is degenerate since it would require $S = \{0\}$, which is not a partition into positive parts. The statement should be: for $k \in \mathbb{Z}_{\geq 1}$, $\mathcal{P}_{\text{special}}((3k^2+k)/2) = \{\{k+1, k+2, \dots, 2k\}\}$. The proof below uses this corrected reading.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\text{special}}(n)$ with base b , slope s , slope set D . By definition, $b \in D$ and either $b = s$ or $b = s+1$.

Sub-step 1a: $S = D$. The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$ and $b = \min(S)$, every element of S lies between b and $\max(S)$. Moreover, the integers $b, b+1, \dots, \max(S)$ all lie in D (because $b \in D$ implies $b \geq \max(S) - s + 1$, so $\{b, b+1, \dots, \max(S)\} \subseteq D$), and $D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b+1, \dots, \max(S)\}.$$

In particular $|S| = \max(S) - b + 1$, and since $S = D$ this also equals s . Thus $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$. Then $|S| = s = b$ and $\max(S) = 2b - 1$, so

$$S = \{b, b+1, \dots, 2b-1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \dots + (b-1)) = b^2 + \frac{b(b-1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2 - k)/2$ and $S = S_{-k}$ (matching the definition $S_{-k} = \{2k-1, \dots, k\}$).

Step 3: Case B — $b = s+1$. Then $|S| = s = b-1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b+1, \dots, 2b-2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b-1) \cdot b + (0+1+\dots+(b-2)) = b(b-1) + \frac{(b-1)(b-2)}{2} = \frac{(b-1)(3b-2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k+1)-2)}{2} = \frac{k(3k+1)}{2} = \frac{3k^2+k}{2}, \quad S = \{k+1, k+2, \dots, 2k\}.$$

Step 4: Forward direction. Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k+1, \dots, 2k-1\}$ has base k , slope k (the entire set is a top-consecutive run), and $b = k \in D$, hence is special. Similarly for the upper family.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n , there is at most one special partition (the candidate is uniquely determined by b). Combined with the forward direction, this gives exactly one.

Step 6: Non-pentagonal n . If n is not pentagonal of either form, no $b \geq 1$ produces $\sum = n$ in the analysis above, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. $\square$

Definition 15 (Operations α and β). Let $S \subseteq \{1, \dots, n\}$ be a partition of n into different parts with base b and slope s and slope set D . We define the following operations on S :

α : If $S \in \mathcal{P}_\alpha$ (equivalently, if $b \leq s$ and $b \notin D$, or if $b \leq s-1$), then we move b up and turn its points into new slope.

$$\alpha(S) = (S \setminus \{b, \max(S) - b + 1\}) \cup \{\max(S) + 1\}.$$

β : If $S \in \mathcal{P}_\beta$ (equivalently, if $b > s$ and $b \notin D$, or if $b \geq s+2$), then we move the slope down and turn its points into the new base.

$$\beta(S) = S \cup \{s, \max(S) - s\} \setminus \{\max(S)\}.$$

α



β



Lemma 16. *If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max $m = \max(S)$, slope set $D = \{m - s + 1, \dots, m\}$. Recall that the α -condition says $b \leq s$ (with $b \notin D$ if $b = s$), and

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

(Equivalently in the more granular form used in the Lean code: remove the base b , remove the bottom-most element of the top- b slope $\{m - b + 1\}$, and add $m + 1$. The remaining $b - 1$ slope elements stay where they are; the smallest one of them is $m - b + 2$.)

Step 1: $\alpha(S)$ is a partition of n . The change in sum is $-b - (m - b + 1) + (m + 1) = 0$, so the sum is preserved. Since b and $m - b + 1$ both lie in S and $m + 1 \notin S$ (as $m = \max(S)$), the operation produces a set, and all parts are positive (the smallest possible new value is $m + 1 \geq 2$, and any retained part is in S , hence $\geq b \geq 1$).

We must verify that $b \neq m - b + 1$ (so we are removing two distinct elements). Equivalently $m \neq 2b - 1$. If $m = 2b - 1$, then $S \supseteq \{b, b + 1, \dots, 2b - 1\} = D$, forcing $S = D$ (since $|S| \geq |D| = s = b$ and $S = \{b, \dots, 2b - 1\}$). But then S would be the special partition S_{-b} , contradicting $S \in \mathcal{P}_\alpha$.

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m + 1$.
- New base: $b' := \min(\alpha(S))$. Since b was removed, b' is the second-smallest element of S (if $|S| \geq 2$) or $m + 1$ (if $|S| = 1$).

Step 3: New slope s' and slope set D' . The slope of $\alpha(S)$ is the maximal consecutive run from m' downward. The retained top- $(b - 1)$ slope elements $\{m - b + 2, \dots, m\}$ together with the new element $m + 1$ form the consecutive run $\{m - b + 2, m - b + 3, \dots, m, m + 1\}$ of length b . We claim this is the entire new slope, i.e., $s' = b$ and $D' = \{m - b + 2, \dots, m + 1\}$.

Sub-step 3a: The run does not extend further down. The next candidate would be $m - b + 1$, which we just removed; so $m - b + 1 \notin \alpha(S)$, terminating the run.

Step 4: Verify $\alpha(S) \in \mathcal{P}_\beta(n)$. We need $b' > s'$ (and $b' \notin D'$) or $b' \geq s' + 2$. We have $s' = b$ from Step 3, and we examine which clause of $\mathcal{P}_\alpha$ produced S :

Sub-case (i): $b \leq s - 1$ (the second clause of $\mathcal{P}_\alpha$). The slope of S has length $s \geq b + 1$, so its top b elements $\{m - b + 1, \dots, m\}$ are a strict subset of D . The element $m - s + 1 \in D$ satisfies $m - s + 1 < m - b + 1$, so $m - s + 1$ is in S and below the part removed. After applying α , this element remains, and its value $m - s + 1$ may or may not be the new minimum.

TODO: Case analysis on the location of the new base b' relative to D' . The careful analysis (matching the Lean lemma `alpha_gives_beta_crit`) shows: the new base $b' \geq b + 1$ (since the second-smallest element of S is $\geq b + 1$; in fact $\geq b + 1$ because elements are distinct), and one verifies that $b' \geq s' + 2$ or $b' > s' \wedge b' \notin D'$. The detailed bookkeeping requires distinguishing whether the second-smallest of S is in the slope or below it.

*Sub-case (ii): $b = s$ and $b \notin D$ (first clause of $\mathcal{P}_\alpha$). **TODO: Symmetric analysis. The conclusion is again that the resulting partition satisfies the β -criterion.*** $\square$

Lemma 17. *If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set D . Recall the β -condition: $b > s$ (with $b \notin D$ if $b = s + 1$), and

$$\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}.$$

(In the granular form: remove the top-of-slope m , shift the slope $D = \{m - s + 1, \dots, m\}$ down by one to $\{m - s, \dots, m - 1\}$, and adjoin a new bottom element $\{s\}$. The retained slope elements stay at $\{m - s + 1, \dots, m - 1\}$; the only "new" element below them is $m - s$, and the only "new" element overall is s .)

Step 1: $\beta(S)$ is a partition of n . Sum change: shifting s slope elements each down by 1 contributes $-s$, and adjoining the new singleton $\{s\}$ contributes $+s$. The net change is zero.

We must verify the resulting set has distinct elements:

- $s \notin S \setminus \{m\}$: since $b > s$ (from β -criterion) and $b = \min(S)$, all parts of S exceed s . So s is genuinely new.
- $m - s \notin S \setminus \{m\}$: if $m - s$ were already in S , then since $m - s < m - s + 1$ (the slope minimum), $m - s$ would be in $S \setminus D$. But by maximality of the slope (the slope is the maximal top-consecutive run; cf. `slope_is_interval` in Lean), if $m - s \in S$ then $\{m - s, m - s + 1, \dots, m\} \subseteq S$, which would mean $m - s \in D$, contradiction.
- $s \neq m - s$, i.e., $m \neq 2s$. **TODO: verify this from the β -criterion. Heuristically, if $m = 2s$ then $D = \{s + 1, \dots, 2s\}$ has minimum $s + 1$; combined with β -criterion ($b \geq s + 2$ in the second clause, or $b = s + 1, b \notin D$ in the first), one checks the case $m = 2s, b = s + 1$ would give $b \in D$ (since $b = s + 1 \in D$), excluded.**
- Positivity: $s \geq 1$ and $m - s \geq 1$ (since $m - s \geq m - s + 1 - 1$ and the slope minimum is ≥ 1).

Step 2: New extremes.

- New base: $b' := \min(\beta(S)) = \min(b, s) = s$ (since $b > s$).
- New maximum: $m' := \max(\beta(S))$. If $|S| \geq 2$, then $m' =$ second-largest element of S , which equals $m - 1$ (the next slope element). If $|S| = 1$, then $S = \{m\} = \{b\}$, but $b > s \geq 1$ means $s \geq 1$, so $|S| = 1$ forces a degenerate case; **TODO: verify $|S| \geq 2$ holds whenever $S \in \mathcal{P}_\beta$, by a small computation.**

Step 3: New slope s' and slope set D' . The shifted slope occupies $\{m-s, m-s+1, \dots, m-1\}$, a consecutive run of length s ending at $m' = m-1$. So the new slope is at least this run. We claim it is exactly this run, i.e., $s' = s$ and $D' = \{m-s, \dots, m-1\}$.

Reason it does not extend further down: the next candidate is $m-s-1$. If this were in $\beta(S)$, it would have to come from $S \setminus \{m\}$ (since neither shift-target nor s equals $m-s-1$ generically). But $m-s-1 \in S$ would, by slope maximality applied to S , require $m-s-1 \in D$, contradicting $\min(D) = m-s+1$.

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$. We have $b' = s$ and $s' = s$, so $b' = s'$. We need either $b' \notin D'$ (giving the first clause of $\mathcal{P}_\alpha$) or $b' \leq s' - 1$ (the second clause). Since $b' = s'$, the second clause fails, and we must show $b' = s \notin D'$.

$D' = \{m-s, m-s+1, \dots, m-1\}$. We have $s \in D'$ iff $m-s \leq s \leq m-1$, iff $m \geq s+1$ (always true since $m \geq b > s$) and $s \leq m-1$, the latter equivalent to $m \geq s+1$.

So generically $s \in D'$ would hold, which would block $\mathcal{P}_\alpha$. But we have additional information from the β -criterion:

- If $b \geq s+2$ (second clause of β): then $m \geq b \geq s+2$, so $m-s \geq 2 > s$ would require... **TODO: this analysis is delicate and the careful statement is that $s \notin D'$, which translates to $s < m-s$, i.e., $m > 2s$. This holds in both clauses of $\mathcal{P}_\beta$, but needs the same kind of casework as in the α direction. See Lean's `beta_gives_alpha_crit` (currently sorry-ed).**

□

Lemma 18. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\alpha(S)$. By the definition of α :

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma ??, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \dots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \dots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition of β :

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \setminus \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \setminus \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned} \beta(\alpha(S)) &= [(S \setminus \{b, m-b+1\}) \cup \{m+1\}] \cup \{b, m-b+1\} \setminus \{m+1\} \\ &= [(S \setminus \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}] \setminus \{m+1\} \\ &= (S \setminus \{b, m-b+1\}) \cup \{b, m-b+1\} \\ &= S, \end{aligned}$$

where we used that b and $m - b + 1$ are in S (so re-adding them after removal recovers S), and that $m + 1 \notin S \setminus \{b, m - b + 1\}$ (since $m + 1 > m \geq$ every element of S). $\square$

Lemma 19. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}.$$

By Lemma ??, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma, the data of $\beta(S)$ is:

- new max $m^\beta = m - 1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m - s, \dots, m - 1\}$),
- new slope set $D^\beta = \{m - s, \dots, m - 1\}$.

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = (\beta(S) \setminus \{b^\beta, m^\beta - b^\beta + 1\}) \cup \{m^\beta + 1\} = (\beta(S) \setminus \{s, m - s\}) \cup \{m\}.$$

(We used $b^\beta = s$, $m^\beta = m - 1$, so $m^\beta - b^\beta + 1 = m - 1 - s + 1 = m - s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

$$\begin{aligned} \alpha(\beta(S)) &= [((S \cup \{s, m - s\}) \setminus \{m\}) \setminus \{s, m - s\}] \cup \{m\} \\ &= [(S \setminus \{m\}) \setminus \{s, m - s\}] \cup \{m\}. \end{aligned}$$

Since $s \notin S$ and $m - s \notin S$ (verified in the proof of Lemma ??), removing them from $S \setminus \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S \setminus \{m\}) \cup \{m\} = S,$$

since $m \in S$. $\square$

Lemma 20. *The map $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ is a bijection.*

The map $\beta : \mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$ is a bijection.

Proof. Lemma ?? shows α is well-defined as a map $\mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$, and Lemma ?? shows β is well-defined as a map $\mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$.

Lemma ?? gives $\beta \circ \alpha = \text{id}_{\mathcal{P}_\alpha(n)}$, and Lemma ?? gives $\alpha \circ \beta = \text{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. $\square$

Lemma 21.

$$|\mathcal{P}_\alpha(n)| = |\mathcal{P}_\beta(n)|.$$

Proof. A bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ from Lemma ?? . $\square$

Lemma 22. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$.*

Proof. Membership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas ?? and ??. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}$:

- we remove two distinct elements (b and $m - b + 1$, distinct as shown in the proof of Lemma ??), reducing the size by 2;
- we add one element ($m + 1$), which is not in $S \setminus \{b, m - b + 1\}$ since $m + 1 > \max(S)$.

Net change: $-2 + 1 = -1$. So $|\alpha(S)| = |S| - 1$, flipping parity.

Step 2: $|\beta(S)| = |S| + 1$. By the definition $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

- we add two distinct new elements s and $m - s$ (distinct, and not in S , as verified in the proof of Lemma ??), increasing the size by 2;
- we remove one element ($m \in S$), reducing by 1.

Net change: $+2 - 1 = +1$. So $|\beta(S)| = |S| + 1$, flipping parity.

Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S| - 1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S| - 1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S| + 1$.

$\square$

Lemma 23.

$$\begin{aligned} |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)|, \\ |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)|. \end{aligned}$$

Lemma 24.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 \pm k}{2} \\ 0, & \text{otherwise.} \end{cases}$$

Proof. We have

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \quad (1)$$

By Lemma ?? we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|.$$

It follows from two previous equations and Lemma ?? that

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)| - |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|. \quad (2)$$

Finally, the statement of the lemma follows from the equations (??), (??) and Lemma ??.

3 Jacobi triple product

Theorem 25. For all $w \neq 0$ and q with $|q| < 1$:

$$\prod_{k=1}^{\infty} (1 - q^{2k}) (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k w^{2k} q^{k^2}.$$

3.1 Proof

Set $\phi_n(w, q) := \prod_{k=1}^n (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2})$. We have

$$\phi_n(qw, q) = \phi_n(w, q) \frac{(1 - q^{2n+1} w^2) (1 - q^{-1} w^{-2})}{(1 - qw^2) (1 - q^{2n-1} w^{-2})} = \phi_n(w, q) \frac{1 - q^{2n+1} w^2}{-qw^2 + q^{2n}}. \quad (3)$$

We consider the following expansion

$$\phi_n(w, q) = \sum_{k=-n}^n A_{n,k}(q) w^{2k}, \quad (4)$$

where $A_{n,k}(q)$ is a polynomial in q . We have a symmetry $A_{n,k}(q) = A_{n,-k}(q)$. We find

$$A_{n,n}(q) = q^{1+3+5+\dots+(2n-1)} = q^{n^2}. \quad (5)$$

We substitute (??) into (??) and obtain

$$A_{n,k}(q) = A_{n,k-1}(q) q^{2k-1} \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}. \quad (6)$$

From (??) and (??) we find

$$A_{n,k} = \frac{q^{k^2}}{(1-q^2) \cdots (1-q^{2n})} \prod_{s=n-k+1}^n (1-q^{2s}) \prod_{s=n+k+1}^{2n} (1-q^{2s}). \quad (7)$$

From (??) we obtain

$$\prod_{k=1}^n (1-q^{2k}) (1-q^{2k-1} w^2) (1-q^{2k-1} w^{-2}) = \sum_{k=-n}^n (-1)^k w^{2k} q^{k^2} B_{n,k}(q).$$

where

$$B_{n,k}(q) = \prod_{s=n-k+1}^n (1-q^{2s}) \prod_{s=n+k+1}^{2n} (1-q^{2s}).$$

We have $B_{n,k} = 1 + O(q^{n-k+1})$. We take the limit $n \rightarrow \infty$ and finish the proof.